BOUNDED ABOVE OBJECTS WITH SHIFT-RELATIVELY-PSEUDO-COHERENT COHOMOLOGY ARE RELATIVELY PSEUDO-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17097-17103 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17105-17109, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. With L16191 this closes the complex-vs-cohomology pair for the relative notion.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohomology, lemma cohomology-lemma-cohomology-pseudo-coherent; chart definition of relative pseudo-coherence.

Source reference. Original lemma 09V8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17109–17109. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a morphism of schemes which is locally of finite type. Let $m \in \mathbf{Z}$. Let E be an object of $D(\mathcal{O}_X)$. If E is (locally) bounded above and $H^i(E)$ is $(m-i)$-pseudo-coherent relative to S for all i, then E is m-pseudo-coherent relative to S.*

Proof. Follows from Cohomology, Lemma 09V8 and the definitions. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.